Communication Network Design Lab
(course of Prof. Massimo Tornatore)

Lab 1

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Lab slides: courtesy of Dr. Omran Ayoub and Dr. Leila Askari

Outline

- Flow Formulation (FF)
  - Splittable flow formulation
    - Introduction to Net2Plan (short demo)
  - Unsplittable flow formulation (homework)
  - Task #1 description
- Route Formulation (RF) without wavelength continuity
- Exercise: Route Formulation (RF) vs. Flow Formulation (FF)

Download Lab-1 material
In WeBeep: Lab Material → 01-LAB-CND-1

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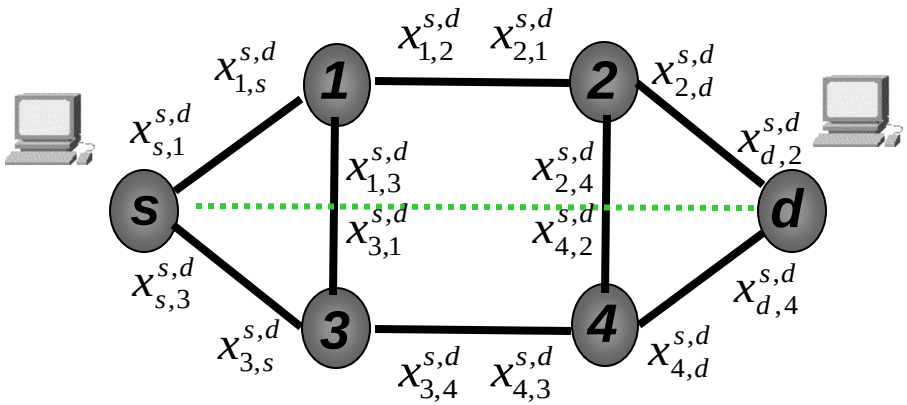
Flow Formulation (Splittable Flow)

Graph G(N,E)	*N set of Nodes, E set of Links*
c	*node pair (source s_c, destination d_c) having requested connection request(s)*
l,k	*link identifiers (source-destination pairs)*
V_c	*number of connection requests having s_c as source and d_c as destination*
x_{lkc}	(integer) number of connections of *c* routed on link *l,k*
W	*Link capacity, number of wavelengths*



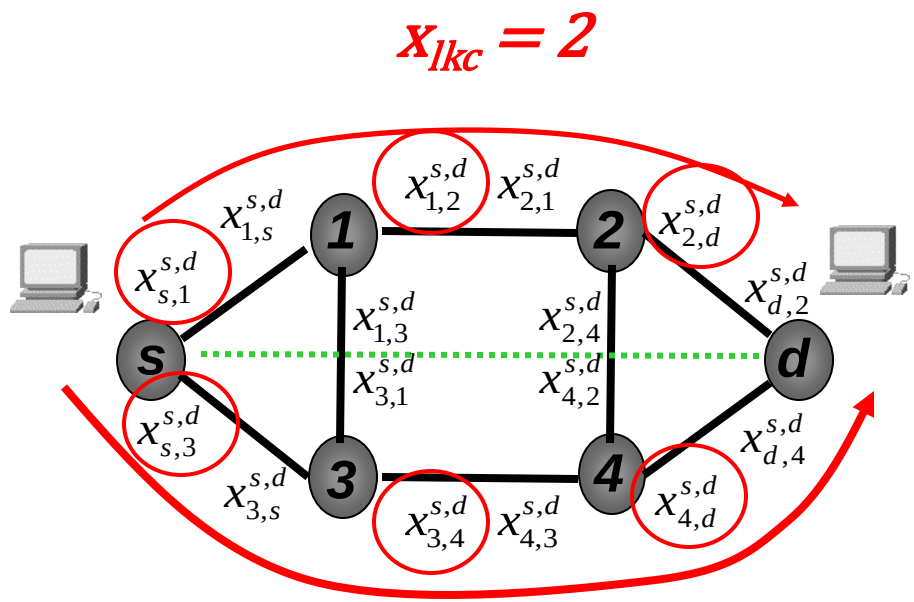
Flow Formulation (Splittable Flow)

- Routing of demands uses ‘flow-link’ decision variables:

x_{lkc} = (integer) number of connections of c routed on link l,k

$$x_{lkc} \geq 0$$

Example, assume V_c ($V^{s,d}$) = 3



V_c number of connection requests having s_c as source and d_c as destination

Flow Formulation (Splittable Flow)

- ***Objective function**:* minimize total amount of traffic routed on all links

$$\min \sum_{c \in C, lk \in E} x_{lkc}$$

- ***Subject to:***

Solenoidal ity	$\sum_{k \in A_l} x_{k,l,c} - \sum_{k \in A_l} x_{l,k,c} = \begin{cases} v_c & \text{if } l = d_c \\ -v_c & \text{if } l = s_c \\ 0 & \text{otherwise} \end{cases} \quad \forall\, l, c$
Capacity	$\sum_{c} x_{l,k,c} \le W \cdot F_{l,k} \quad \forall (l,k)$
Integrity	$x_{l,k,c}$ integer $\quad \forall c,(l,k)$ $F_{l,k}$ integer $\quad \forall (l,k)$

Number of fibers on link l,k

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Anyone that has not yet downloaded the software tools?

Open 'Net2plan-Demo-CND-LAB-1.pdf'

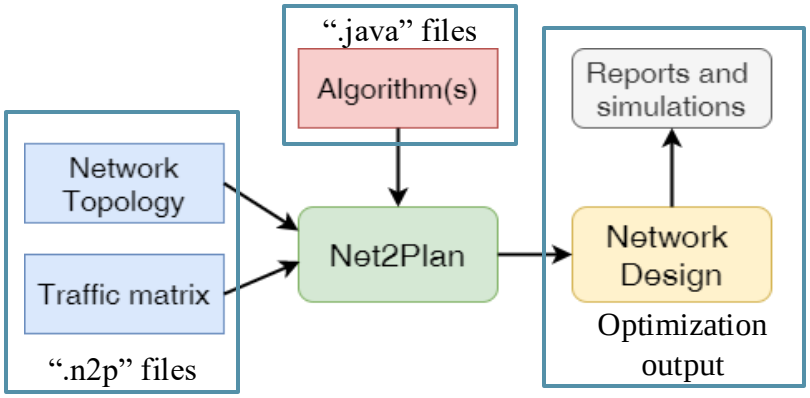
Up to part-B of the document

Flow Formulation (Splittable Flow)

- Now we want to compile and inspect the java code of *splittable flow formulation* to use in Net2Plan (**Part B of Net2Plan demo**)

- **SplittableFlowFormulation.java** can be found in the **'01 - Code'** folder **in OneDrive (01 – CND-LAB-1 material)** or in **WeBeep (Lab Material: 01-LAB-CND-1)**

Link Capacity (Gbps)	*Objective Function*	*Time*
10		
15		
20		
30		
50		



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Flow Formulation (Unsplittable Flow)

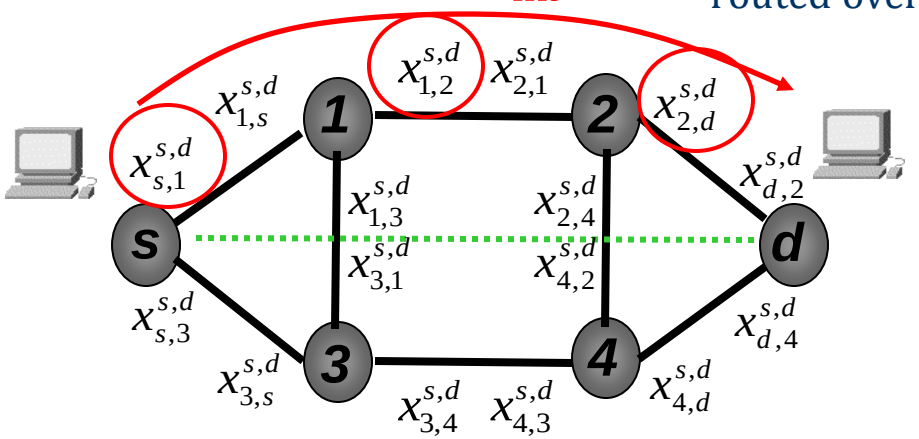
- Routing of demands uses 'flow-link' decision variables
- $x_{lkc} = \{0,1\}$ if offered traffic of c is routed on link l,k

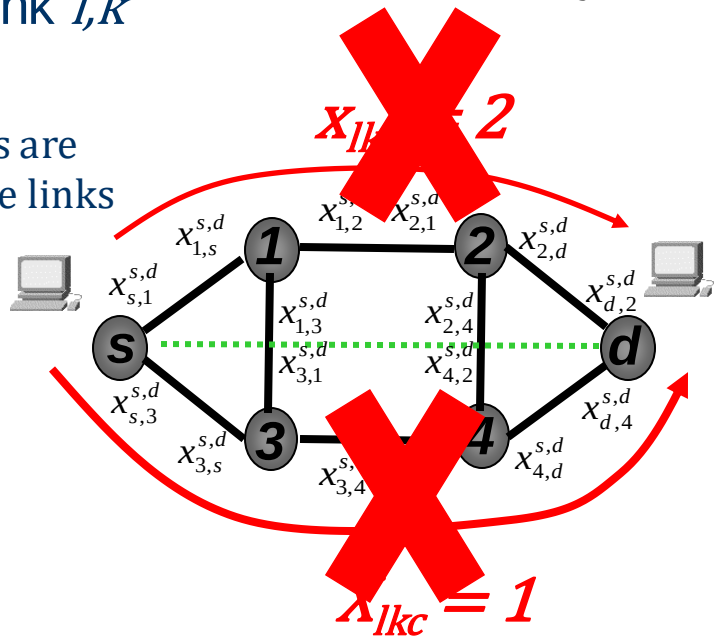
Decision variable is now binary

Example, assume V_c ($V^{s,d}$) = 3

$x_{lkc} = 1$

All 3 connections are routed over these links





- ***Objective function***: *minimize total amount of traffic routed on all links*

$$\min \sum_{c \in C, lk \in E} V_c \, x_{lkc}$$

Objective function changes with respect to splittable FF

V_c *number of connection requests having s_c as source and d_c as destination*

Flow Formulation (Unsplittable Flow)

- **Subject to:**

Decision variable is now binary, flow constraint → 1 and not Vc

$$\sum_{k \in Al} x_{klc} - \sum_{k \in Al} x_{lkc} = \begin{cases} 1 & l = d_c \\ -1 & l = s_c \\ 0 & Otherwise \end{cases} \qquad \forall\ l, c$$

$$\sum_{c \in C} \mathrm{V_c}\, \mathbf{x_{lkc}} \leq WF_{lk} \qquad \forall\ l, k$$

Capacity constraint also changes. Decision variable of demand c on link l,k * connections of demand c <= link capacity l,k

$$\mathrm{x_{lkc}} \text{ is binary} \qquad \forall\ l, k, c$$

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Task #1

- Modify ***FlowFormulation.java*** to assume <u>unsplittable flows</u>
  - What are the differences with respect to splittable FF?
    - Check slides #10 and #11

- Load the 7-nodes network topology with traffic (the network topology used during the demo)
  - Compare '*splittable*' & '*unsplittable*' varying link capacity $C = \{10,15,17,20,30,40,50\}$
  - Plot/tabulate total carried traffic with respect to C
  - Please refer to the task template submission form
    - Submit pdf file and code via e-mail with subject "***LAB_CND_task_#***" to qiaolun.zhang@polimi.it and memedhe.ibrahimi@polimi.it

Task #1: Hints for task completion

1. Include “HashSet” and “Set” java libraries

   “import java.util.HashSet;”

   “import java.util.Set;”

2. Important modification to the code for unsplittable FF
   - Comment line-64 and uncomment line-65 and line-66
3. Modify objective function according to equations given in the slides
4. Modify solenoidality and capacity constraints according to equations discussed in the slides

Link Capacity (Gbps)	*Splittable*	*Unsplittable*
10		
15		
17		
20		
30		
40		
50		

Values of objective function for splittable and unsplittable FF as a function of link capacity

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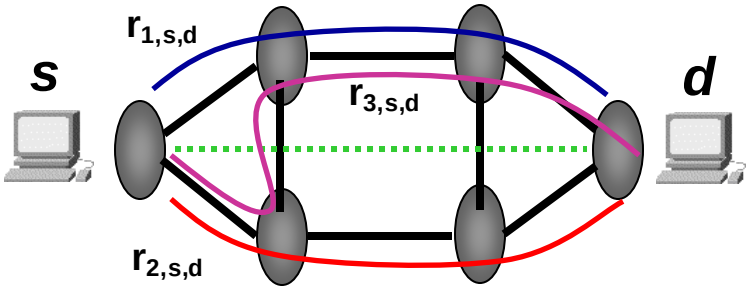
RF without wavelength continuity

- (All) possible paths between each (*s, d)* -pair are evaluated *a priori*
  - Constrained routing (k shortest paths)
- Variables represent which path is used for a given connection
  - r_{cn}: path *n* is used by r_{cn} connections between s_c and d_c

G(N,E)	*N set of Nodes, E set of Links*
c	*node pair (s_c and d_c)*
l,k	*link identifiers (source-destination pairs)*
n	*path identifier*
V_c	*number of connections requests having s_c as source and d_c as destination*
$r_{c,n}$	*(integer) number of connections routed on n-th admissible path for node-couple c*
W	*Link capacity, number of wavelengths*
P_n	*Number of links on path n*



RF without wavelength continuity

- ***Objective function***: *minimize total amount of capacity used (capacity consumption) to route demands on all links*

$$\min \sum_{c,n} r_{cn} \cdot P_n$$

Wavelength is not accounted for:
no wavelength continuity

Solenoidal ity	$\sum_{n} r_{c,n} = v_c \quad \forall c$
Capacity	$\sum_{r_{c,n} \in R_{l,k}} r_{c,n} \leq W \cdot F_{l,k} \quad \forall (l,k)$
Integrity	$r_{c,n}$ integer $\quad \forall (c,n)$ $F_{l,k}$ integer $\quad \forall (l,k)$

$R_{(l,k)}$: set of (all) admissible paths passing through link (l,k)

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RF without wavelength continuity

- **Exercise: Route Formulation (RF) vs. Flow Formulation (FF)**
- File: '*RouteFormulationVWP.java*'
- Found in WeBeep: Lab Material → LAB-1 material

- According to Part-B of Demo:
  1) Copy '*RouteFormulationVWP.java*' to Eclipse workspace in '*src*' folder of the project you created during Lab-1
  2) In 'Eclipse' you find and open '*RouteFormulationVWP.java*' and click '*Ctrl +S*' '*RouteFormulationVWP.class*' file is generated in '*bin*' folder of the project
  3) We can now import the '*RouteFormulationVWP.class*' in Net2Plan

Flow Formulation (FF) vs. Route Formulation (RF)

- SplittableFlowFormulation.java (LAB-1, we compiled and got '.class' file)
- RouteFormulationVWP.java (LAB-2: compile it to get '*.class*' file)
  - Assume k (number of shortest paths calculated a-priori) = {1, 2, 3}

- *Load 'example7nodes_withTraffic.n2p' (set C = 25, 30, 35)*
- Compare FF and RF in terms of capacity consumption and execution time in seconds

Formulation	k	C = 25	C = 30	C = 35	Time (sec.)
FF	NA				
RF	1				
	2				
	3				

Flow Formulation (FF) vs. Route Formulation (RF)

	k	(4) C = ?	(2) C = ?	(3) C = ?	C = 25	(1) C = ?	C = 30	C = 35	Time (s)
FF	NA								
RF	1								
	2								
	3								

- (1)-(4): follow these additional steps to find the values of C for which there is no solution when k = 1, 2, 3 in RF
  - (1): What value of C ensures a solution in case of k = 1
  - (2): What value of C leads to having no solution for k = 2
  - (3): Objective function values for C values between (2) and C = 25
  - (4): is there a value of C for which there is no solution also in case k = 3